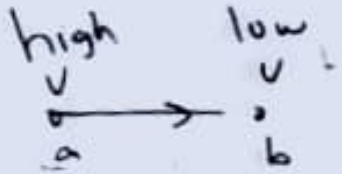


Ohm's law

Aim:-

[1] verifying Ohm's law experimentally

[2] Determination of unknown resistance (R).

$$V = \frac{W}{Q} = \boxed{\frac{J}{C} = V}$$

$r_v = \infty$

$$V = \boxed{I} R$$

$$I \quad \frac{Q}{t} = \frac{ne}{t} \rightarrow 1.6 \times 10^{-19} C = \boxed{\frac{C}{s} = A}$$

Ohm's law

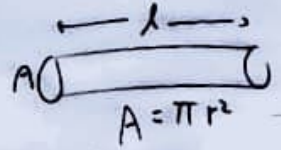
$$\boxed{V = IR}$$

Aim:-

[1] verifying ohm's law experimentally

[2] Determination of unknown resistance (R).

$$R(\Omega) = \frac{\rho l}{A}$$

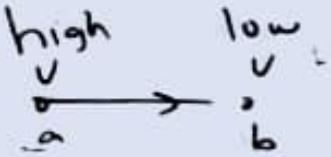


$$\boxed{R \propto T} \rightarrow$$ Conductor (Cu, Fe, Al, ...)

$$\boxed{R \propto \frac{1}{T}} \rightarrow$$ Semiconductor (Si, Ge, ...

$$\rho = \frac{RA}{l} = \frac{\Omega m^2}{m} = \Omega m$$

$$\sigma = \frac{1}{\rho} = \Omega^{-1} \cdot m^{-1}$$



Ohm's law

Aim:-

[1] verifying ohm's law experimentally

[2] Determination of unknown resistance (R).

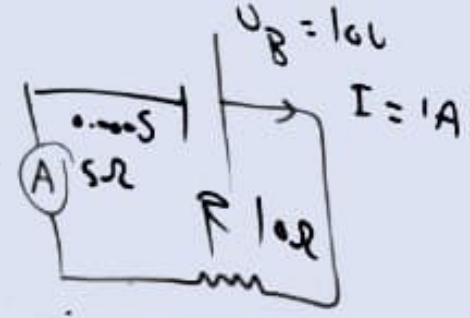
$$V = \frac{W}{Q} = \left[\frac{J}{C}\ V\right]$$

$r = \infty$

$$V = [I]R$$

$$I = \frac{Q}{t} = \frac{ne}{t} \quad (ne \rightarrow 1.6\times10^{-19}\ C) = \left[\frac{C}{s} = A\right]$$

$r = 0$



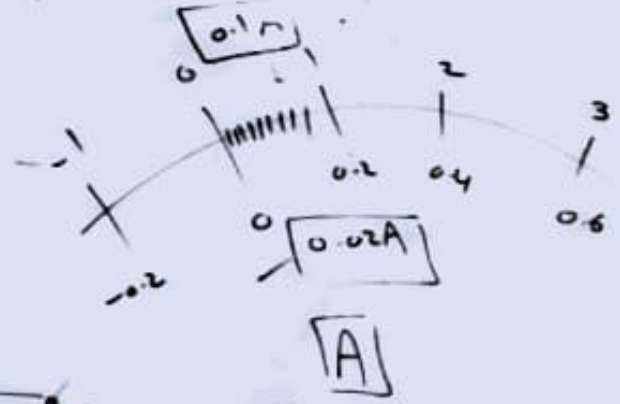
- الهدف
- النتائج
- الأسئلة
- الخطوات
- النظرية
- حل الحرارة

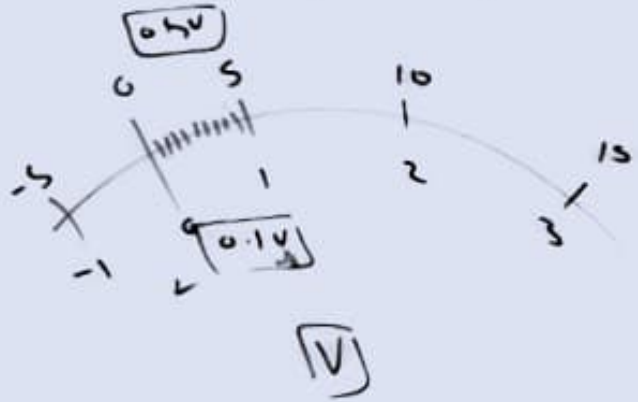
30
10 لكل
10
10

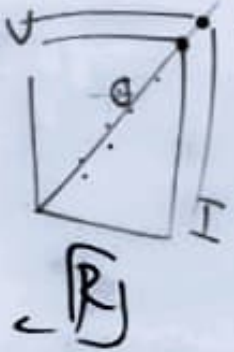
70

	Current	Voltage	Resistance
سير تجربة			
النتائج			
الوحدة الكاملة			
الجهاز			

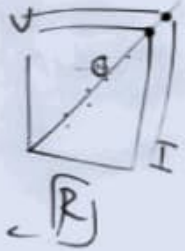
V(V)	I(A)

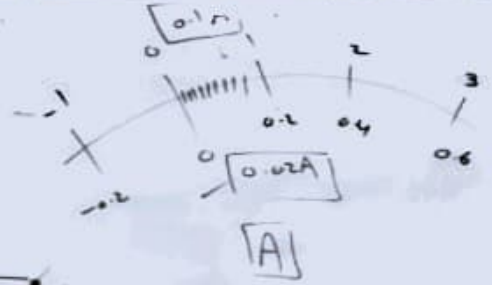


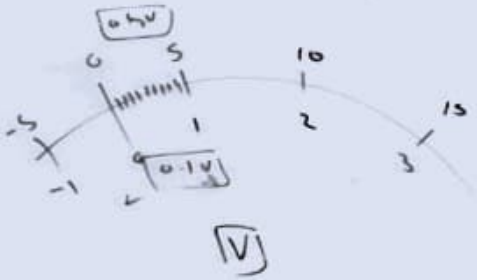




V(V)	I(A)







Aim: [1] achievement ohm's Law Experimentally

(2) Determin an unknown Resistance

law $R = \frac{V}{I}$

[1] Current intensity

[2] voltage (Potential)

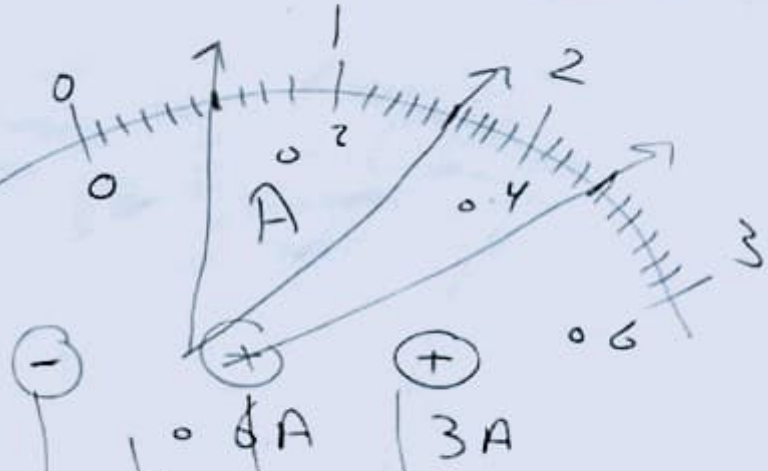
[3] Resistance

[4] ohm's law $V \propto I$

$V = R \cdot I$

$R = \frac{V}{I}$

5 ohm's Law



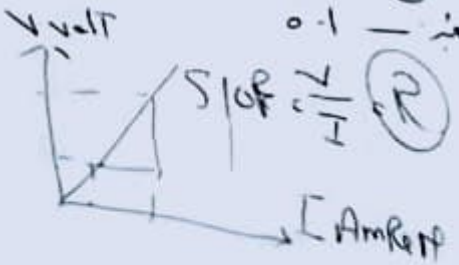
(1) لو كان بتوصيل مع 1.5 ابر قيمة المقاومة

يكام = 5 Ω

(1) لو انا عايز اعرف 0.6 ، 1 الرقم المقاومة يبقى

0.02

0.1 الرقم المقاومة ← 3A (2)

الحل: $R = \frac{V}{I}$ Slope

I	V